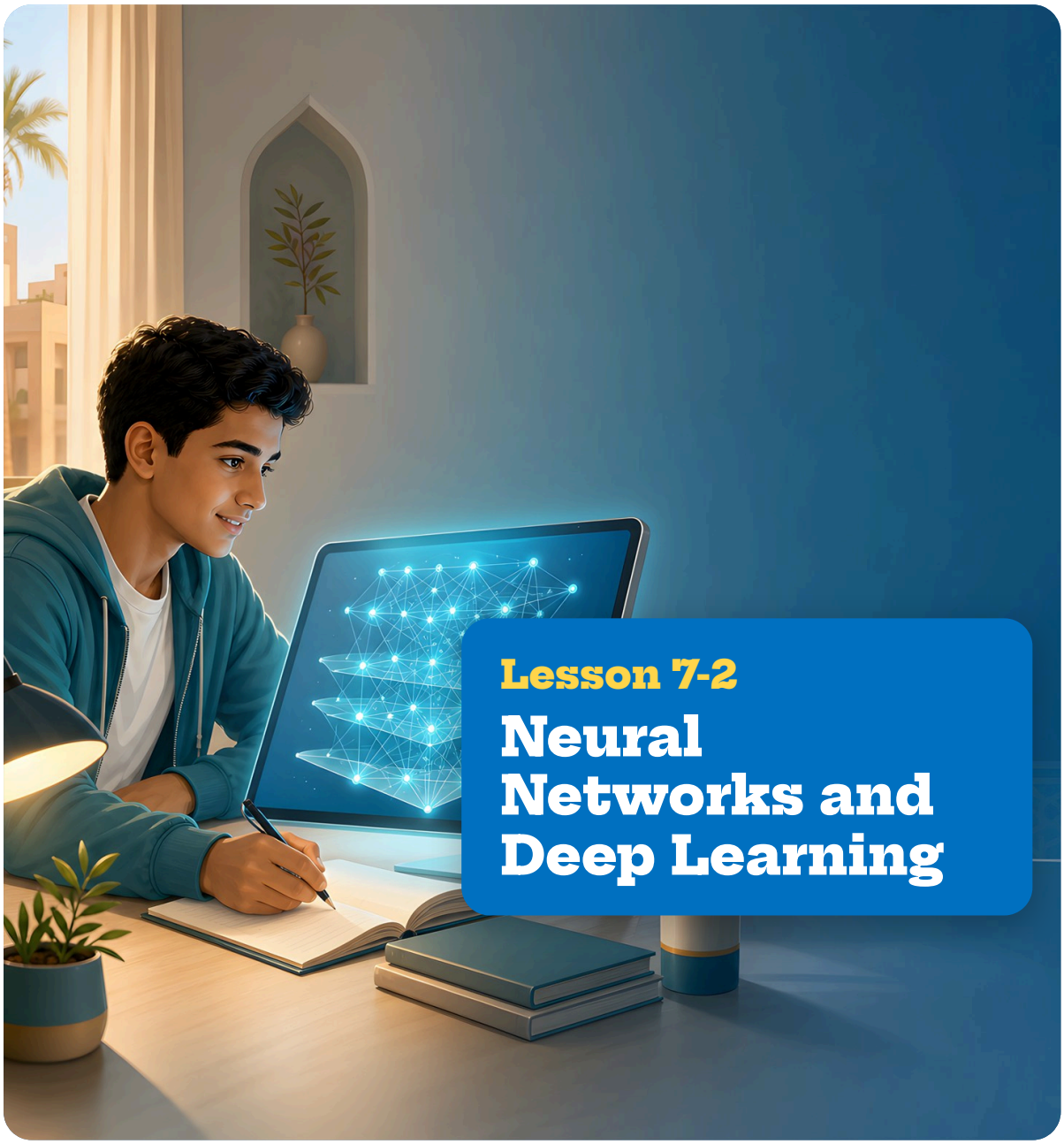


Lesson 7-2

Neural Networks and Deep Learning

Learning Objectives

- Describe the structure of a neural network and explain how a single neuron processes its inputs.
- Explain what deep learning is, and why stacking many hidden layers brought a leap in AI performance.

How can a machine tell a dog from a cat, or turn speech into text? The answer, loosely inspired by the brain, is a neural network: layers of simple units called neurons, each passing signals to the next. A neuron just multiplies its inputs by weights and adds them up — nothing complex on its own — but stack enough of them across enough layers and the network learns, by adjusting those weights on data, to recognize patterns no human could write rules for. Making the stack deep — deep learning — lets the network build up features from simple edges up to whole objects, and it is what pushed AI past humans on many tasks.

? Guiding question: How is a neural network built and trained, and what does making it "deep" add?

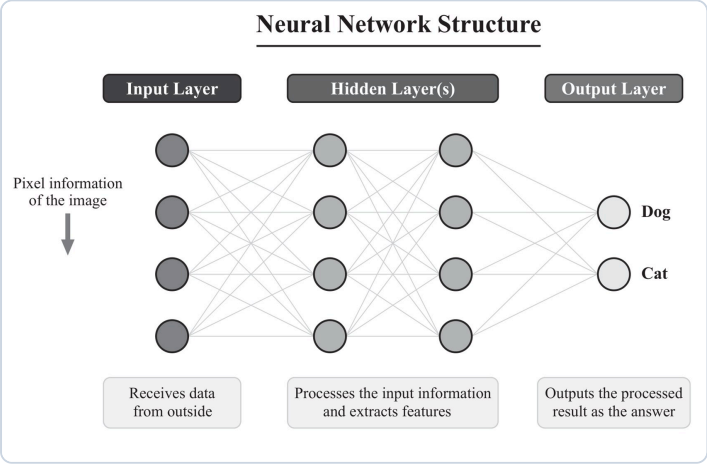
Point!

1. The Structure of Neural Networks

(1) A neural network has a structure in which many **(nodes)**, also called neurons, are connected. Nodes are arranged together in **(layers)**, and the nodes of adjacent layers are connected to each other.

(2) Layers are divided by role into the following three types.

Type of layer	Role
Input layer	Receives data from outside
Hidden layer (middle layer)	Processes the input information and extracts features
Output layer	Outputs the result of processing as an answer



A neural network: input layer, hidden layers, and output layer

(3) Data enters at the input layer, is processed as it passes through the hidden layers, and reaches the output layer in this flow. *e.g.,* In a network that judges whether the animal in a photograph is a dog or a cat, the input layer receives the

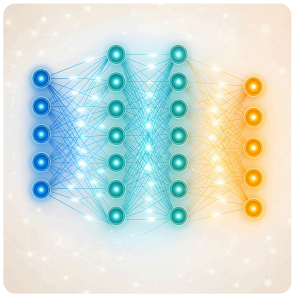
Key Fact:
A neural network is layers of connected neurons: an input layer, hidden layers that extract features, and an output layer. Each neuron multiplies its inputs by weights and sums them; training adjusts those weights from data. Deep learning stacks many hidden layers to build features step by step — the leap behind modern AI.

Key Concepts:

- node (neuron), layer
- input / hidden / output layer
- weight, training
- deep learning
- hierarchical features

KEY TERMS

node (neuron) — a single processing unit.
layer — a group of neurons at one stage.
input / hidden / output layer — receive, process, answer.



A network stacks neurons into connected layers.

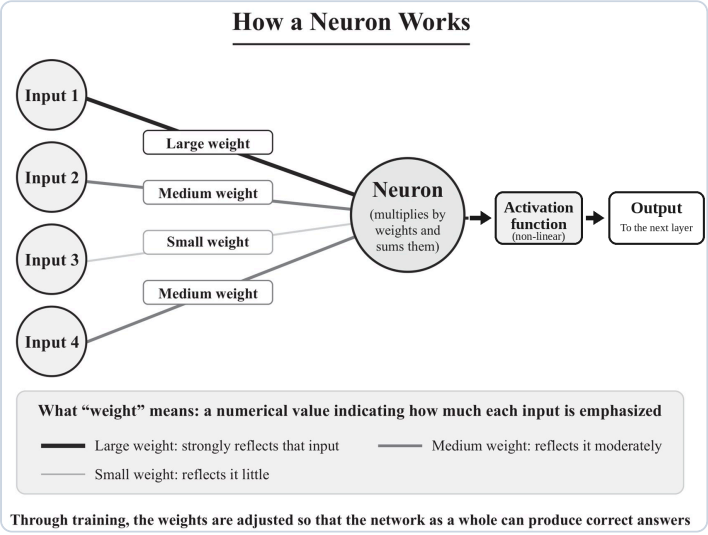
pixel information of the photograph, the hidden layers extract features such as shape and pattern, and the output layer outputs the judgment of “dog” or “cat.”

(4) The number of hidden layers and the number of nodes in each layer are designed according to the difficulty of the problem to be handled. The more complex the problem, the more layers and nodes are needed.

2. How a Neuron Works

(1) Each neuron receives multiple inputs, multiplies them by weights and sums them, then passes the sum through an **(activation function)**, which lets the network learn relationships that are not simply proportional, before passing the result to the next layer.

(2) **(Weight)**…a numerical value that represents how much each input is emphasized. Weights can be negative as well as positive, and the larger a weight is in size, the more strongly that input influences the neuron's result.



A neuron weights its inputs, sums them, applies an activation function, and passes the result on

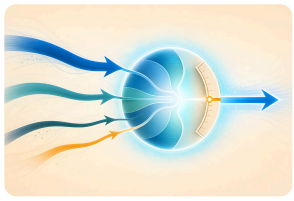
(3) **(Training)** is the process of gradually adjusting the weights between neurons using a large amount of data.

Stage	What happens
Before training	The weights are set to arbitrary values, and correct predictions cannot be made
During training	Data is given, and the weights are gradually adjusted based on the difference between the prediction and the correct answer
After training	With the adjusted weights, correct predictions can be made even on new data

(4) As training progresses, a combination of weights that allows the network as a whole to produce correct answers is found. Even though the function of each individual neuron is simple, complex judgments become possible by combining many neurons.

KEY TERMS

weight — how much an input is emphasized.
training — adjusting weights from data.



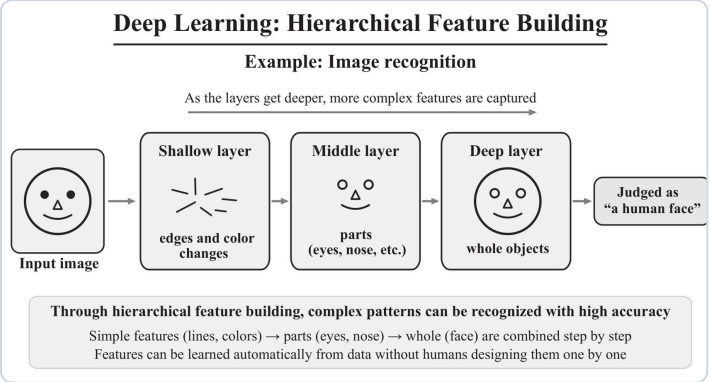
A neuron weights its inputs, sums them, applies an activation function, and passes the result on.

3. Deep Learning

(1) **(Deep learning)** is a technology that uses neural networks with many hidden layers stacked deeply. “Deep” refers to having many hidden layers.

(2) When the hidden layers are made deeper, features of data can be built up step by step and captured. *e.g.,* In an image-recognition network, features are built up from shallow to deep layers as follows.

Layer position	Examples of features extracted
Shallow layers	Simple features such as lines and changes in color
Middle layers	Shapes of parts such as eyes, nose, and ears
Deep layers	Structure of the whole face or object



Deep learning builds features step by step, from edges to whole objects

(3) Through this hierarchical building of features, deep learning can recognize complex patterns with high accuracy.

(4) Deep learning became practical as a result of large amounts of data and high computational power (in particular, specialized processors well suited to AI computation, such as GPUs and TPUs) becoming available. Before that, it was technically difficult to advance learning sufficiently even when layers were made deeper.

4. Why Deep Learning Brought a Leap in AI

(1) With the availability of deep learning, accuracy close to or surpassing humans has become achievable in many tasks that were previously considered difficult for computers.

(2) The fields in which there has been particularly great progress include the following.

PAUSE & THINK

Why does stacking more hidden layers let a network capture more complex features?



Deep layers build from edges to whole objects.

PAUSE & THINK

What two things had to become available before deep learning could work in practice?

Field	Examples
(Image recognition)	Face recognition systems, diagnostic support for medical images, object detection for self-driving cars
(Speech recognition)	Voice operation of smart speakers, transcription from speech, speech translation
(Natural language processing)	Machine translation, summarization of text, answering questions

(3) Before deep learning, features had to be designed by humans one by one and given to the computer. With deep learning, features can be learned automatically from data, allowing the limitations of human design to be exceeded.

(4) As a result of accumulating these capabilities, the foundation has been laid for AI that newly creates text and images.

✎ Worked Example

(1) Mark each statement (A–D) with ○ if true or × if false.

A A neural network has three types of layers: input layer, hidden layer, and output layer.

B The input layer is the layer that outputs the results of processing as an answer to the outside.

C Deep learning is a technology that uses a neural network with only one hidden layer.

D Training a neural network is the process of adjusting the weights between neurons using a large amount of data.

(2) Match each description (a–c) with the most appropriate term from the choices below (A–C).

a In a neural network, the layer that receives data from outside

b In a neural network, a numerical value that represents how much each input is emphasized

c In a neural network, the layer that processes input information and extracts features

[Choices] **A** Hidden layer **B** Weight **C** Input layer

✓ Solution

(1)

A There are three types of layers: input, hidden, and output. Therefore, ○.

B The output layer is what outputs results to the outside. The input layer is the layer that receives data from outside. Therefore, ×.

C Deep learning is a technology that uses neural networks with many hidden layers stacked deeply. Therefore, ×.

PAUSE & THINK

Which layer receives the data, and which one gives the final answer?

D Correct description of training. Therefore, ○.

(2)

a: The one that receives data is the input layer. **C**

b: The value representing how much an input is emphasized is the weight. **B**

c: The one that processes information and extracts features is the hidden layer. **A**

Try

1 Answer the following questions.

1. What is the term for the individual processing unit that makes up a neural network?
2. In a neural network, what is the term for the layer that processes input information and extracts features?
3. In a neural network, what is the term for the numerical value that represents how much each input is emphasized?
4. What is the term for the technology that uses neural networks with many hidden layers stacked deeply?

2 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes the layers of a neural network.

A The input layer has the role of outputting processing results to the outside.

B The hidden layer has the role of receiving data from outside.

C The output layer has the role of processing input information and extracting features.

D Data enters at the input layer, is processed as it passes through the hidden layers, and reaches the output layer in this flow.

(2) From the following options (A–D), choose the one that most appropriately describes the training of a neural network.

A From before training begins, the weights between neurons are set to correct values.

B As training progresses, a combination of weights that allows the network as a whole to produce correct answers is found.

C Training is performed by increasing the number of neurons.

D Once training is complete, the network becomes unable to handle new data.

EXAM-STYLE QUESTION

Explain how a neural network with simple neurons can make complex judgments, referring to weights and training. **[6]**
Refer to: weighting inputs and adjusting weights on data.

PAUSE & THINK

Before training, are the weights already correct? What changes them?

(3) From the following options (A–D), choose the one that is NOT an appropriate characteristic of deep learning.

A It has a structure with many hidden layers stacked deeply.

B It can build up features of data step by step.

C The deeper the hidden layers, the more complex features can be captured.

D Even if the number of hidden layers is increased, the difficulty of problems that can be handled does not change.

(4) For a deep learning network that performs image recognition, reorder the following (A–C) in the order in which features are extracted from shallow layers to deep layers.

A Structure of the whole face or object

B Simple features such as lines and changes in color

C Shapes of parts such as eyes, nose, and ears

PAUSE & THINK

More hidden layers let a network capture more complex features — but what does that extra depth demand in data and computing power?

Think as an Engineer — Investigate & Decide

A team wants an app that reads handwritten digits 0–9 from scanned school registers and turns them into typed numbers. You advise on building it with a deep learning network.

☐ **Write the rules for each digit by hand**

☐ **Train a deep network on many examples**

1 · Design the layers. Say what the input layer would receive, what the output layer would produce, and in one line why hidden layers (rather than a single step) are needed here.

2 · Explain the depth. Using the idea of hierarchical features, describe what shallow, middle, and deep layers might each capture for a handwritten digit — a chain the lesson shows for faces but does not spell out for digits, so you must map it yourself.

3 · Judge the cost and risk. Name the two things this network needs to become practical, and name one way the training data could be too narrow, making the app fail on some students' handwriting. **In pairs,** compare designs and agree on the one dataset change that would most improve fairness.

Give evidence for your answer.

Stuck? Use the lesson's face example as a model, then translate it to a digit: ask what would count as an “edge,” a “part,” and a “whole” for a written number. For the risk, list whose handwriting the network saw during training — then ask whose it never saw.

KEY TERMS

training data — the labeled examples a network learns from.

generalize — work correctly on new, unseen data.

🌍 In a New Context

A voice assistant turns spoken commands into text and often works well even for words it was not explicitly programmed with. Name which field of AI progress this belongs to, and explain, using layers and training, how it can handle speech no one wrote rules for.

❓ Lesson Question — Answered

The lesson asked how a neural network is built and trained and what making it deep adds. A neural network is built from layers of connected neurons: an input layer that receives data, hidden layers that process it and extract features, and an output layer that gives the answer, with data flowing input → hidden → output. Each neuron is simple — it multiplies its inputs by weights, sums them, and passes the sum through an activation function — and the network is trained by gradually adjusting those weights on large amounts of data. Before training, arbitrary weights give wrong answers; after training, tuned weights give correct ones even on new data. Making the network deep — deep learning, many hidden layers — lets features be built hierarchically, from simple edges through parts to whole objects, so complex patterns are recognized with high accuracy. Once large data and powerful processors like GPUs made it practical, deep learning let features be learned automatically instead of designed by hand, driving leaps in image recognition, speech recognition, and natural language processing.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Read the following passage and answer each question.

A neural network has a structure in which many processing units called (a) are connected. These are arranged together in (b), and are divided into three types: (c), which receives data from outside; (d), which processes information and extracts features; and (e), which outputs the result as an answer. Each (a) multiplies its inputs by (f), sums them, passes the sum through an activation function, and passes the result to the next layer. Training gradually adjusts (f) using large amounts of data. The technology that uses neural networks with many hidden layers stacked deeply is called (g), and its characteristic is that it can build up features of data (h) and capture them.

⭐ REFLECT & CHALLENGE

Reflect: Where have you used a tool built on deep learning (voice typing, photo search, translation)? Which field was it?
Challenge: A team makes a network deeper and deeper but accuracy stops improving and training becomes very slow. Using what made deep learning practical, explain what their deeper network now demands more of, and name the one thing — besides adding layers — they should check first.

1. Fill in the blanks (a)–(h).

3 Answer the following questions.

(1) From the following options (A–D), choose the one that most appropriately describes how a neuron in a neural network works.

A Each neuron passes one input as is to the next layer.

B Each neuron multiplies its inputs by weights, sums them, passes the sum through an activation function, and passes the result on.

C Each neuron averages the input values and passes the result to the next layer.

D Each neuron passes only the largest input to the next layer.

(2) From the following options (A–D), choose the one that most appropriately describes the background to deep learning becoming practical.

A Computation became simpler by reducing the number of hidden layers to one.

B Large amounts of data and high computational power became available.

C Connections between neurons were eliminated, simplifying processing.

D A method in which humans set weights one by one by hand was established.

(3) From the following options (A–D), choose the one that most appropriately describes why deep learning brought a leap in AI.

A Features could be learned automatically from data without humans designing them one by one.

B Computers became able to think with the same mechanism as the human brain.

C Computers became able to fully understand the meaning of words.

D Computers became able to make completely accurate judgments on any problem.

Key takeaway

Remember:
A neural network = layers of neurons (input → hidden → output); each neuron weights and sums its inputs, and training tunes the weights on data. Deep learning stacks many hidden layers to build features (edges → parts → wholes), enabled by big data and GPUs.